

5.3.11

AI25BTECH11017-BALU

Question:

Find $\mathbf{A}^2 - 5\mathbf{A} + 6\mathbf{I}$ if

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix} \quad (0.1)$$

Solution:

Let us solve the given equation theoretically and then verify the solution computationally
According to the question,

We know that

$$\|\mathbf{A} - \lambda\mathbf{I}\| = 0 \quad (0.2)$$

$$\left\| \begin{pmatrix} 2-\lambda & 0 & 1 \\ 2 & 1-\lambda & 3 \\ 1 & -1 & -\lambda \end{pmatrix} \right\| = 0 \quad (0.3)$$

$$\lambda^3 - 3\lambda^2 + 4\lambda - 3 = 0 \quad (0.4)$$

using Cayley-Hamilton theorem

$$\mathbf{A}^3 - 3\mathbf{A}^2 + 4\mathbf{A} - 3\mathbf{I} = 0 \quad (0.5)$$

$$\mathbf{A}^2 - 3\mathbf{A} + 4\mathbf{A} = 3\mathbf{A}^{-1} \quad (0.6)$$

$$\mathbf{A}^2 - 5\mathbf{A} + 6\mathbf{I} = 3\mathbf{A}^{-1} - 2\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{pmatrix} \quad (0.7)$$